# LAB3 Report

Pin-Jing, Li (111511015 *ouo.ee11@nycu.edu.tw)*
Ching-Kai, Huang (111511151 *kevin.ee11@nycu.edu.tw)*
Duan-Kai, Hu (112511015 *huduankai@gmail.com)*

November 9, 2025

# 1 Data Pre-Processing

# 2 Experiment I: Car movement

## 2.1 Calibration

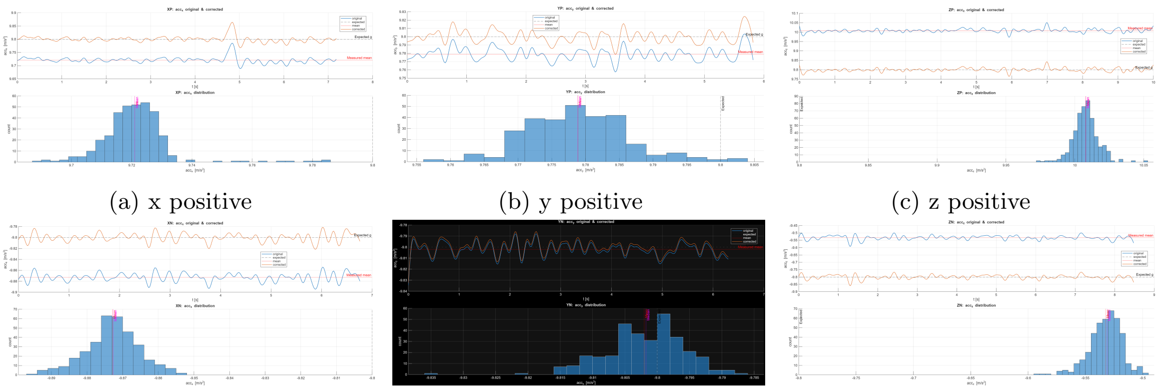
This section talks about the calibration on the accelleration detector of the IMU. We make use of the gravity $1g = 16384$ as our reference. The work flow is as follow,

1 Face the target direction upward and collect the signal.

2 Obtain the mean of the target direction in raw unit with `meas_mean = mean(target_axis)`

3 Obtain the desired correction term `b_axis = a_exp - meas_mean` with `a_exp = 16384`

The following set of figures shows the signal before and after the calibration.



(a) x positive (b) y positive (c) z positive

(d) x negative (e) y negative (f) z negative

Figure 1: Calibration of accelleration on six different direction

## 2.2 Car Displacement

# 3 Experiment II: Walking Motion Detection

## 3.1 Calibration

Since what we're interesting in is the total displacement on the three direction, in this section we directly pass the signal through a low pass filter.

## 3.2 Coordinate System Transformation

## 3.3 Walking Motion Analysis

## 3.4 Walking Trace Reconstruction

## References